

Default Project Proposal

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Abstract

This proposal outlines a comprehensive project to reproduce, fine-tune, and extend the GPT-2 architecture toward multilingual natural language inference (NLI). The project is divided into four stages: implementing GPT-2 & the Adam optimizer from first principles & validating the implementation through a toy pre-training task, fine-tuning the pretrained GPT-2 on the English NLI dataset (XNLI), extending the task to 15 languages to evaluate zero-shot & multilingual transfer capabilities, and conducting open-ended experiments to explore efficiency, performance, or interpretability. This work aims to deepen the understanding of transformer-based language models, assess GPT-2's cross-lingual generalization, and explore directions for lightweight multilingual adaptation.

1 Introduction

The transformer-based architecture has made significant breakthroughs in natural language processing, reducing the need to depend on crucial yet less effective architectures such as LSTMs. Among them, GPT-2 introduced a scalable, autoregressive pre-training framework that enables strong few-shot performance across diverse tasks. Despite its success, GPT-2 was pre-trained primarily on English data, leaving its multilingual capabilities under-explored. This project aims to replicate GPT-2's architecture and training mechanisms from scratch to gain a fundamental understanding of its operation and to extend its use to multilingual NLI tasks.

Natural Language Inference (NLI) is a core task in language understanding. It determines whether a hypothesis can be logically inferred from a premise. By formulating NLI as a text generation problem, predicting label tokens using next-token prediction, this project fine-tunes GPT-2 directly on linguistic entailment relationships. Extending this setup to 15 languages provides an opportunity to analyze GPT-2's implicit cross-lingual representations and tokenizer efficiency.

The broader goals are:

- to reproduce GPT-2's internal components for educational insight
- to empirically study its adaptability to new languages
- to propose practical improvements for multilingual fine-tuning and evaluation as extensions.

2 Related Work

Pre-trained Language Models, such as GPT-2 by [Radford et al. \(2019\)](#), transformed the NLP field by proposing self-supervised training approach as the key, establishing foundation for general purpose language models.

To consider Natural Language Inference (NLI), The XNLI benchmark by [Conneau et al. \(2018\)](#) evaluates cross-lingual understanding through entailment, contradiction, and neutrality relationships. Previous work ([Hu et al., 2020](#)) has shown that

monolingual models have limited transfer performance to unseen languages, highlighting the need for multilingual fine-tuning strategies.

Multilingual adaptation has been explored in WECHSEL by [Minixhofer et al. \(2022\)](#), which uses subword embeddings reinitialization to transfer monolingual models such as GPT-2 into other languages with lower compute.

Due to the massive increase in the sizes of the above discussed models, different size reduction techniques have been proposed to speed up the inference and improve usability. Hence, different size compression techniques have been proposed:

- GPTQ, by [Frantar et al. \(2023\)](#), enabling post-training quantization of Transformer weights to 3-4 bits without substantial loss in accuracy
- QuAdapter, by [Park et al. \(2022\)](#), serving as an efficient post-training quantization technique and improves quantization performance

At the same time, adaptation techniques that reduce the cost of the fine tuning of transformers emerged. LoRA, proposed by [Hu et al. \(2021\)](#), introduces low-rank matrices that adapt model weights without the need to update the full parameter set. Additionally, MAD-X by [Pfeiffer et al. \(2020\)](#) is an adapter based technique that introduces the idea of modular layers that can be inserted for cross lingual transfer tasks. The above techniques enable reusing base models such as GPT-2 for different specific tasks with reduced compute and storage overhead.

3 Methodology

3.1 Task 1: Implement GPT-2

The group shall implement the missing core functions to a small GPT-2 architecture. These functions include causal self-attention, transformer blocks and embedding/output layers. Besides this, the group will implement AdamW optimizer with bias-corrected moment to estimates and decoupled weight decay, then train the GPT-2 toy model with a small text corpus by using these components with the objective being prediction of the next-token. Training loss will be monitored through iterations & epochs and compare against provided sanity checks. Ultimately, the goal of Task 1 is to have verified & working implementations of GPT-2 and AdamW optimizer.

3.2 Task 2: English NLI with GPT-2

The group shall use a pre-trained GPT-2 model for NLI in English to classify premise-hypothesis relationships as entailment, contradiction, or neutral. An official pre-trained GPT-2 weights from HuggingFace would be loaded. There would be an implementation of a greedy text generation function that repeatedly builds attention masks and runs GPT-2. It will also convert the last hidden state to logits and selects the next token using argmax.

Then, the XNLI dataset for English NLI will be used via its `XNLIDataset` class. It handles TSV loading, tokenization, padding, batching, and label mapping. The team shall implement the training loop with batch processing, loss computation, backpropagation, and optimizer updates. Then the group will evaluate model performance on the development set after each epoch, save the best-performing checkpoint, conduct hyperparameter exploration by adjusting learning rate, batch size, or weight decay and analyze their effects on model performance. By the end, the final fine-tuned model will be evaluated on the test set using accuracy as the primary metric.

3.3 Task 3: Multilingual NLI with GPT-2

The objective is to evaluate and improve GPT-2 on NLI in multiple languages. First, by testing how well English-only training transfers to other languages. Then, by analyzing how the tokenizer handles words in different languages and comparing fine-tuned GPT-2 model on NLI in all those languages. After performing zero-shot cross-lingual transfer by evaluating the English fine-tuned model from Task 2 on all 14 non-English languages in XNLI, the group shall select target languages based on combined insights from zero-shot performance and fertility scores, prioritizing languages with lower fertility and reasonable zero-shot accuracy. Finally, the team will analyze which languages benefit most from fine-tuning and investigate potential correlations between linguistic properties and model performance

3.4 Task 4: Open-ended Explorations

The final stage of the project focuses on experiments to increase our comprehension of the model and of the architecture used in GPT-2

1. Visualizing Attention Mechanisms

The group could develop visualization tools to

interpret GPT-2's internal representations, especially attention, as it is a major feature of the transformer architecture. Attention heatmaps will be generated to display how tokens attend to one another across different layers and heads. By comparing attention patterns between languages and label types (entailment, contradiction, neutrality), one could explore whether certain linguistic features, such as negation or syntactic alignment, are being captured, and whether it is possible to visualize attention patterns.

2. Architectural and Parameters Modification

To assess efficiency–performance trade-offs, the team can experiment with modifying GPT-2's architecture. This includes parameter reduction and/or architectural variations. Taking the former as an example, the team may reduce the number of layers (encoder/decoder), attention heads, and other parameters to observe the impact on model performance, training and inference speed. As this is trained with a limited amount of data, it may be interesting to see if fewer parameters could help with the training.

3. Data Scaling

The group may plan to extend the fine-tuning dataset to explore the relationship between dataset size and performance. Additional NLI data will be incorporated to test data efficiency and generalization effects. If the dataset scale exceeds local hardware limits, we consider buying some cloud GPU resources to enable larger-scale training. After the initial evaluation, the team may re-run the parameter modification experiments under these larger data conditions to test how previous experiments result evolved with data size.

4 Expected Outcomes and Evaluation

After completing all main tasks, the GPT-2 model is expected to demonstrate correct implementation and learning behaviour across its stages. During the pre-training experiment, the main goal is to verify that the model has been properly built and that it can learn from the data. We expect to observe a decrease in training loss over time. When fine-tuned on the Natural Language Inference (NLI) task, the model should achieve a level of performance above

random guessing (i.e., better than 33% accuracy across the three labels: entailment, contradiction, and neutral). This indicates that the model has learnt to capture some level of semantic and logical structure from the data. In the multilingual extension, performance is expected to drop for non-English languages but to remain interpretable and better than random guessing.

5 Timeline

This proposal will be due on Sunday, November 9 (end of Week 8). By then, the first task would already be underway. In Week 9, the focus will be on completing Task 1 and 2. Afterwards in Week 10, the focus is on completing Task 3 and then on the open-ended extension, whereby the team will work on the tasks of visualising attention mechanisms and start on architectural & parameter modifications. Should this complete by Week 11, the final report and slides will be submitted by the end of Week 12, while our team gets ready for the final project presentation on Week 13.

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